

TASK- 1

1. INTRODUCTION SECTION

Embedded Systems: An **embedded system** is a specialized computer system designed to perform a specific function or a set of functions within a larger device. It combines hardware and software and is usually dedicated to a particular task.

An embedded system mainly consists of two parts: hardware and software. The hardware includes components such as a microcontroller or microprocessor, memory, sensors, actuators, and input/output interfaces. The software is usually a program written in languages like C or C++ that controls the hardware and allows the system to perform its specific function.

Real World Applications: Embedded systems are specialized computing systems designed to perform dedicated functions within larger devices. They have become an essential part of modern technology and are widely used across various industries.

1. Consumer Electronics

Embedded systems are found in devices such as washing machines, microwave ovens, digital cameras, smart televisions, and air conditioners. They control device operations and improve user experience.

2. Automotive Industry

Modern vehicles contain numerous embedded systems, including engine control units (ECUs), anti-lock braking systems (ABS), airbag controllers, cruise control systems, and parking assistance systems. These systems enhance vehicle performance and safety.

3. Healthcare and Medical Devices

Medical equipment such as pacemakers, ECG monitors, infusion pumps, MRI scanners, and glucose monitoring systems rely on embedded systems for accurate monitoring and control.

4. Industrial Automation

Factories use embedded systems in programmable logic controllers (PLCs), robotic arms, CNC machines, and process control systems to improve productivity, precision, and efficiency.

5. Communication Systems

Smartphones, routers, modems, network switches, and satellite communication devices use embedded systems to manage data transmission and communication processes.

6. Home Automation and Smart Devices

Embedded systems are the backbone of smart homes, enabling devices such as smart locks, security cameras, smart lighting systems, and smart thermostats to operate intelligently.

7. Aerospace and Defense

Aircraft navigation systems, flight control systems, radar equipment, and missile guidance systems use embedded technology to ensure reliability and real-time operation.

8. Banking and Retail

Automated Teller Machines (ATMs), Point-of-Sale (POS) terminals, barcode scanners, and vending machines employ embedded systems for secure and efficient transactions.

Difference between microcontroller and microprocessor : Microcontrollers and microprocessors are fundamental components of modern electronic systems. Although both are programmable devices used for processing data and controlling operations, they differ significantly in architecture, functionality, and applications. Microcontrollers and microprocessors are fundamental components of modern electronic systems. Although both are programmable devices used for processing data and controlling operations, they differ significantly in architecture, functionality, and applications.

2 . COMPONENT STUDY

Role Of Microcontroller : A microcontroller is a compact integrated circuit designed to control specific tasks in embedded systems. It contains a processor (CPU), memory, and input/output (I/O) peripherals on a single chip. Microcontrollers are widely used in automation, consumer electronics, robotics, healthcare devices, and Internet of Things (IoT) applications.

one widely used microcontroller is Arduino. Arduino is an open-source microcontroller platform designed for beginners, students, and hobbyists. It provides an easy-to-use programming environment and extensive hardware support.

ESP32 is a powerful microcontroller developed by Espressif Systems. It includes built-in Wi-Fi and Bluetooth connectivity, making it highly suitable for IoT applications.

Raspberry Pi Pico is a low-cost microcontroller board based on the RP2040 microcontroller developed by the Raspberry Pi Foundation.

Basic Sensors : Sensors are important components in embedded systems because they help devices detect changes in the environment. A sensor collects information such as temperature, light, or movement and converts it into an electrical signal that a microcontroller can understand. After processing the signal, the system can take the required action.

A temperature sensor is used to measure the temperature of the surrounding environment. It converts temperature changes into signals that can be read by a microcontroller. Temperature sensors are commonly used in air conditioners, refrigerators, weather monitoring systems, and industrial machines to maintain proper temperature level.

A motion sensor is used to detect movement in a specific area. It works by sensing changes in infrared radiation or physical movement around it. Motion sensors are widely used in security systems, automatic lighting systems, and smart home devices. For example, lights can turn on automatically when a person enters a room.

A light sensor is used to measure the amount of light in an environment. It detects light intensity and sends the information to a microcontroller. Light sensors are commonly used in automatic street lights, mobile phone brightness control, solar tracking systems, and smart lighting system

Documentation & Report

MOTION DETECTION ALARM

INTRODUCTION

A Motion Detection Alarm is a security and automation system that detects movement within a specific area and activates an alarm, buzzer, or LED. It is commonly used in homes, offices, industries, and smart security systems to monitor unauthorized movement.

Components required: Motion Sensor (PIR Sensor), Microcontroller (Arduino, ESP32, or Raspberry Pi Pico), Buzzer, LED Indicator, Power Supply.

Working Principle

The motion detection alarm primarily works using a Passive Infrared (PIR) sensor. Every living being emits infrared (IR) radiation due to body heat. The PIR sensor continuously monitors the infrared radiation in its surroundings.

Step-by-Step Operation

Monitoring Stage

The PIR sensor continuously detects infrared radiation from the environment. Under normal conditions, the infrared pattern remains relatively constant.

Motion Detection

When a person or animal moves within the sensor's range, the infrared radiation pattern changes.

The PIR sensor detects this change and generates an electrical signal.

Signal Processing

The electrical signal is sent to the microcontroller.

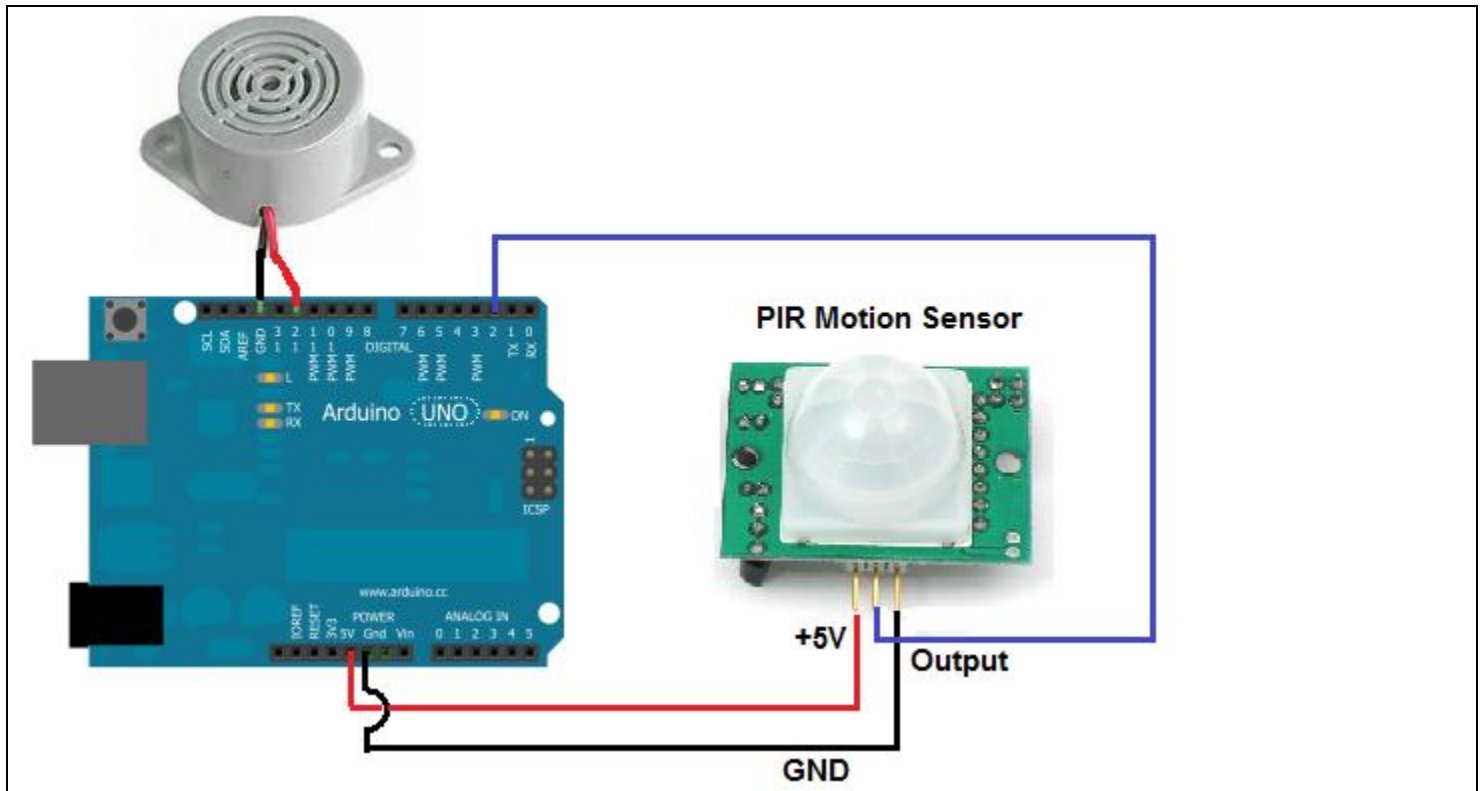
The microcontroller analyzes the signal and determines whether motion has occurred.

Alarm Activation

If motion is detected, the microcontroller activates the buzzer or alarm.

Additional actions such as turning on lights, sending notifications, or recording video can also be triggered.

CIRCUIT DIAGRAM



SOURCE CODE :

```
// Motion Detection Alarm using PIR Sensor
```

```
const int pirPin = 2;    // PIR sensor output pin
```

```
const int ledPin = 13;   // LED pin
```

```
const int buzzerPin = 8; // Buzzer pin
```

```
void setup() {
```

```
    pinMode(pirPin, INPUT);
```

```
    pinMode(ledPin, OUTPUT);
```

```
    pinMode(buzzerPin, OUTPUT);
```

```
    Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
    int motion = digitalRead(pirPin);
```

```
    if (motion == HIGH) {
```

```
        digitalWrite(ledPin, HIGH);
```

```
        digitalWrite(buzzerPin, HIGH);
```

```
        Serial.println("Motion Detected!");
```

```
        delay(1000);
```

```
    }
```

```
    else {
```

```
        digitalWrite(ledPin, LOW);
```

```
        digitalWrite(buzzerPin, LOW);
```

```
        Serial.println("No Motion");
```

```
}  
  
}
```

CODE EXPLANATION

1. The PIR sensor continuously monitors movement.
2. When motion is detected, the PIR sensor output becomes **HIGH**.
3. The Arduino reads this signal and:
 - Turns **ON the LED**.
 - Activates the **buzzer**.
 - Displays "**Motion Detected!**" on the Serial Monitor.
4. When no motion is detected, the LED and buzzer remain **OFF**.

Connections

Component	Arduino pin
PIR Sensor VCC	5V
PIR Sensor GND	GND
PIR Sensor OUT	D2
LED (+)	D13
LED (-)	GND (through 220Ω resistor)
Buzzer (+)	D8
Buzzer (-)	GND

Applications of Motion Detection Alarm

Motion detection alarms are widely used in security, safety, and automation systems. They detect movement within a specified area and generate an alert through a buzzer, LED, notification, or other response mechanisms.

1. Home Security

Motion detection alarms are used in homes to detect unauthorized entry and alert occupants of potential intruders.

2. Office and Building Security

They help protect offices, banks, and commercial buildings by monitoring restricted areas and triggering alarms when movement is detected.

3. Smart Lighting Systems

Motion sensors can automatically switch lights ON when a person enters a room and OFF when the area is vacant, saving energy.

4. Industrial Safety

Factories and industries use motion detection systems to monitor hazardous zones and ensure worker safety.

5. Automatic Doors

Shopping malls, hospitals, airports, and hotels use motion sensors to automatically open and close doors when a person approaches.

6. Surveillance Systems

Motion detection alarms are integrated with CCTV cameras to start recording when movement is detected.

7. Parking and Vehicle Security

They are used in vehicle security systems and parking areas to detect unauthorized movement and prevent theft.

8. Smart Home Automation

Motion sensors are used to control appliances, security devices, and other smart home functions automatically.